



Tougu Xiaotong capsules may inhibit p38 MAPK pathway-mediated inflammation: In vivo and in vitro verification

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ABSTRACT

Ethnopharmacological relevance: Tougu Xiaotong capsules (TXC) are an herbal compound commonly used to treat osteoarthritis (OA) in China.

Aim of the study: We attempted to verify TXC's therapeutic effects and mechanisms related to the p38 mitogen-activated protein kinase (MAPK) pathway in vivo and in vitro.

Materials and methods: TXC's therapeutic effects were assessed by observing cartilage degeneration and inflammatory factors in a modified Hulth's model (in vivo) and a lipopolysaccharides (LPS)-exposed cellular model (in vitro). The expression of biomarkers related to p38 MAPK pathway-mediated inflammation was also investigated.

Results: TXC treatment reversed cartilage degeneration related biomarkers (ADAMTS 4, ADAMTS 5, Col I, Col V, MMP 3, MMP 9, and MMP 13) and inflammation factors (IL-1 β , TNF- α , and IL-6) in both the animal and cellular OA models. Expression of p-p38 MAPK was downregulated following TXC administration, and changes to microRNAs in the cellular models were recovered. These results indicated that the p38 MAPK pathway-related mechanism may involve therapeutic effects of TXC.

Conclusions: This study verified TXC's efficacy to treat OA in vivo and in vitro and suggests that p38 MAPK pathway-related mechanisms may be involved in TXC's therapeutic effects.

1. Introduction

Osteoarthritis (OA), a chronic musculoskeletal disease, is characterized by articular cartilage degeneration and subchondral bone remodeling. Since OA reduces quality of life and imposes a socio-economic burden, it is considered a severe public health problem (Ashraf et al., 2018; Kwon et al., 2018). Current OA treatments, which include nonsteroidal anti-inflammatory drugs and opioids, are far from satisfactory as they provide only temporary relief from symptoms (Yazici et al., 2017). Thus, there is an urgent need to develop safe and more effective OA therapies. In this context, alternative

treatments—including Chinese herbal medicines—have been considered.

Tougu Xiaotong capsules (TXC) are a herbal compound containing *Morinda officinalis*, *Paeonia lactiflora* Pall, *Ligusticum wallichii* and *Sarcandra glabra* (Table 1) (License number in Fujian Province of Food and Drug Administration: MINZHIZI Z20100006) and are widely used to treat OA in China (Li et al., 2013, 2014a, 2014b; Liu et al., 2007; Wu et al., 2019; Zheng et al., 2009). A number of clinical studies in Chinese verified the clinical efficacy of TXC in treating OA (Chen and Hong, 2015; Hong et al., 2010; Li et al., 2017). The effective components of TXC are paeoniflorin, ferulic acid, isofraxidin and rosmarinic acid

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List of abbreviations

ADAMTS	A disintegrin and metalloproteinase with thrombospondin motifs
Bcl-2	B-cell lymphoma 2
BMD	bone mineral density
BV/TV	bone volume/total volume
Col II	collagen II
CT	computerized tomography
ECM	extracellular matrix
ELISA	Enzyme linked immunosorbent assay
ER	endoplasmic reticulum
GAPDH	glyceraldehyde-3-phosphate dehydrogenase
HE staining	hematoxylin and eosin staining

HPLC	high-performance liquid chromatography
LPS	IL-1 β interleukin -1 β

lipopolysaccharides

MAPK	mitogen-activated protein kinase
miRNAs	microRNAs
MMP	matrix metalloproteinase
PVDF	polyvinylidene difluoride
qRT-PCR	quantitative reverse transcription
OA	osteoarthritis
TLR	Toll-like receptor
TNF- α	tumor necrosis factor - α
TXC	Tougu Xiaotong capsule

Table 1

The ingredients of the Tougu Xiaotong capsule.

Chinese name	Pinyin (Chinese phonetic alphabet)	English name	Part used	Proportion of ingredients (100%)
巴戟天	Ba Ji Tian	Morinda officinalis	Dried root	33.33%
白芍	Bai Shao	Paeonia lactiflora Pall	Dried root	33.33%
川芎	Chuan Xiong	Ligusticum wallichii	Dried root	16.67%
肿节风	Zhong jie Feng	Sarcandra glabra	Dried root	16.67%

(Chen et al., 2015; Li et al., 2014b). Our previous work demonstrated that TXC upregulated expression of B-cell lymphoma 2 (Bcl-2), down-regulated expression of p53, suppressed Caspase-3 and Caspase-9 activity in chondrocytes (Li et al., 2011), and rearranged subchondral collagen fibers and phosphate crystals in an animal model (Li et al., 2013). We also found that TXC's therapeutic effects may be related to an endoplasmic reticulum (ER) stress pathway (Liang et al., 2018). Recently, we found that expression of inflammatory factors like IL-1 β and TNF- α downregulated by treatment of TXC in a rabbit OA model, which indicated that regulating subchondral bone remodeling through inflammatory factors might play a role in the mechanisms of TXC (Wu et al., 2019). Taken together, these findings suggest that TXC mechanisms are complex and multifold.

OA is closely associated with inflammation-mediated cartilage matrix degeneration, which might relate to the p38 mitogen-activated protein kinase (MAPK) pathway (Chen et al., 2018). The relationship between OA and the p38 MAPK pathway has been reported in several previous studies (Lafont et al., 2016; Scotece et al., 2018; Xu et al., 2017; Zhou et al., 2015). One reported that Zhuanggu Jianxi Decoction, a Chinese herbal compound, inhibited caveolin-1 expression, p38 phosphorylation, interleukin (IL)-1 β , tumor necrosis factor (TNF)- α , and matrix metalloproteinase (MMP) 3 and MMP 13 (Yan et al., 2014). Another study found that activation of the p38 MAPK pathway might stimulate expression of pro-inflammatory cytokines, chemokines, MMPs, and signaling enzymes (COX-2) in human osteoarthritis chondrocytes (Feng et al., 2017). Blockage of p38 MAPK with the p38 inhibitor has been shown to suppress chondrocyte apoptosis, decrease downstream inflammatory cytokine production, and prevent recruitment of other inflammatory cells, which may contribute to degeneration of bone and cartilage (Sun et al., 2017). These findings suggest that Chinese herbs used to treat OA may suppress the p38 MAPK pathway. In a computational pharmacology study, our group showed that TXC inhibits the biological effects of inflammatory cytokines, MMP 3, and oxygen free radicals. Moreover, microRNAs (miRNAs), such as miR-27b, miR-140, miR-92a-3p, and miR-34a, play crucial roles in inflammatory responses by inhibiting the expression of target genes, which may be the important regulator of cartilage inflammation and a new target for OA therapy. We therefore wondered whether TXC exerts its anti-inflammatory effects in OA via the p38 MAPK pathway.

In the present study, we examined TXC's effects on inflammation in

an OA animal model (in vivo) and a lipopolysaccharides (LPS)-exposed cellular model (in vitro) to verify whether TXC can relieve inflammation-mediated cartilage matrix degeneration. These experiments will further verify our previous work (Liang et al., 2018). We also investigated the roles of the p38 MAPK pathway and inflammation-related miRNAs to clarify the mechanisms underlying TXC's therapeutic effects. We attempted to provide evidence for the mechanisms of TXC, which may be helpful for future introducing, verifying and using TXC to treat OA in the mainstream medicine.

2. Methods and materials**2.1. Preparation of TXC extracts**

TXC was prepared as in our previous study (Liang et al., 2018). Briefly, all herbs were dried at 50 °C in an air-circulating oven for 24 h and then crushed in a cutting mill (ZN-400A; Zhongnan Pharmaceutical Machinery Factory, China). TXC was prepared at a ratio of 2:2:1:1 of *Morinda officinalis* (Voucher Number 161101, Fujian Haicheng Medicine Co., Ltd., China); *Paeonia lactiflora* Pall (Voucher Number 160201, Fujian Haicheng Medicine Co., Ltd.); *Ligusticum chuanxiong* Hort (Voucher Number 151001, Anhui Xiehecheng Pharmaceutical Tablet Co., Ltd., China); and *Sarcandrae Herba* (Voucher Number 16121706, Anhui Xiehecheng Pharmaceutical Tablet Co., Ltd.) with distilled water. We assessed quality via a high-performance liquid chromatography (HPLC) fingerprint method using an Agilent 1200 HPLC system (Agilent, Santa Clara, CA, USA) and an Ultimate™ XB-C18 column (4.60 × 250.00 mm, 5 μ m, Welch Materials, Inc., USA). The identified components of TXC (Table 1) were consistent with previous studies (Chen et al., 2015; Liang et al., 2018). The HPLC results are provided in Supplemental Fig. 1 and confirm the high quality of the TXC extract (Supplemental Fig. 1), consistent with our previous study (Liang et al., 2018).

2.2. Establishment of OA animal and cell models, subgrouping, and TXC administration

Sixty male Sprague-Dawley rats aged eight weeks (Shanghai Slack Laboratory Animal Co., China) were randomly assigned to a sham group ($n = 20$) or surgical group ($n = 40$). The animals were cage-

reared at 21°C–22 °C under a 12 h/12 h light/dark cycle with free access to food and sterile tap water. All animals were treated as per the National Institute of Health Guidelines for the Care and Use of Laboratory Animals. All experimental procedures were approved by the Animal Care and Use Committee of the Fujian University of Traditional Chinese Medicine (approval number: [2018] Fuzhongyi Lunli Shenzidi No.020).

After one-week acclimation, the surgical group underwent the modified Hulth's method with isoflurane (1.5–2%) (Chen et al., 2016). Briefly, a 1 cm longitudinal incision was made on the skin of the medial right-posterior knee, the medial collateral and anterior cruciate ligaments were transected via the medial approach, and the medial meniscus was removed. The sham group only received a 1 cm longitudinal incision on the skin of the medial right-posterior knee. The rats were allowed to move and feed freely after two weeks. Three rats died within two weeks of surgery, leaving 19 rats in the sham group and 38 rats in the surgical group. Rats in the surgical group were further randomly divided into the model group ($n = 19$) and TXC group ($n = 19$). The dose of TXC was 184 mg/kg/day similar to our previous study (Li et al., 2013). This dose was calculated according to a method to extrapolate dose of TCM from human to animal (Huang et al., 2004). Briefly, dose of rat was calculated as dose of human (adult) $\times$ weight coefficient. Since the recommended dose of TXC in human (standard BW 60 kg) is 1840 mg/day, we got the dose of rat in the present study is 184 mg/kg ($1840/60 \times 6$). TXC was dissolved in normal saline and administered intragastrically for 12 consecutive weeks. Rats in the sham group received intragastric administration of the same volume of normal saline for the same duration.

In addition, a LPS-exposed cellular model was established using previously described methods (Li et al., 2014a). Eighteen four-week-old male Sprague-Dawley rats were used. All experiments were repeated three times, with six animals/time. While the rats were deeply anesthetized with isoflurane (1.5–2%), chondrocytes were isolated from the knee articular cartilage. Chondrocytes were cultured in low-glucose DMEM (Gibco, USA) supplemented with 10% fetal calf serum (Gibco), penicillin (100 U/mL), and streptomycin (100 μ g/mL). Only second passage chondrocytes were identified by collagen II (Col II) immunohistochemistry. Toll-like receptor (TLR) 4 expression on chondrocytes was detected with immunofluorescence (Supplemental Fig. 2). Chondrocytes were exposed to 10 ng/mL LPS (Sigma-Aldrich, USA) for 8 h to establish the cell model. Cells were exposed to 10 ng/mL LPS for 4, 8, 12, or 24 h. We measured the IL-1 β and TNF- α in the cell supernatants from culture because the anti-inflammatory effects of IL-1 β and TNF- α are conducted only when they were secreted to the cell matrix. Enhancement of IL-1 β and TNF- α in culture solution indicates that we had succeeded in establishment a LPS-exposed cellular model, which also had been verified in our previous study in rabbit OA model (Wu et al., 2019). The levels of IL-1 β and TNF- α in culture solution were significantly higher than those in intact (untreated) cells; the maximum effect was exhibited at 8 h. We thus selected 8 h exposure for the subsequent experimental steps (Supplemental Fig. 3). To investigate the roles of the p38 MAPK pathway, SB203580 (Cell Signaling Technology, Inc., USA), an antagonist of p38 MAPK pathway, was used. Second passage chondrocytes were divided into a control group, a model group (LPS 10 ng/mL), a SB203580 group (10 μ mol/mL + LPS 10 ng/mL), and a TXC group (300 μ g/mL + LPS 10 ng/mL) and underwent treatments for 8 h.

2.3. Tissue preparation

Collection of knee joint tissue from the right hind limb was performed after removing the surrounding soft tissue, including muscles and fat, spanning ± 2 cm of the knee joint. Knee joint tissue was fixed in 4% paraformaldehyde (Leagene Bio., China) for 48 h and then stored in 75% ethanol until use for scanning and staining. Cartilage was collected by gentle cutting with a surgical knife, with caution to avoid

cutting the subchondral bone. Cartilage was then immediately frozen in liquid nitrogen and stored at -70 °C for mRNA and protein detection.

2.4. Micro-computerized tomography (CT) and hematoxylin and eosin (HE) staining

The microarchitecture of the knee joint was analyzed by micro-CT scan (60 KV, Ltd., China) using a Micro View software (3DMed 4.1) to calculate bone mineral density (BMD) and bone volume/total volume (BV/TV). After micro-CT scanning, the knee joint tissue was decalcified in 10% Na₂EDTA (Sigma-Aldrich) (PH 7.40 $\pm$ 0.05) for four weeks, with replacement of the decalcification solution every 2–3 days. The fixed-conventional paraffin embedded tissue was then cut into 8 μ m consecutive slices for standard HE staining (Solarbio, China). Histopathological changes in the cartilage and subchondral bone were observed under an optical microscope.

2.5. Array rat extracellular matrix (ECM) and adhesion molecules assay

Three samples of total RNA were examined from femoral condyles of the sham, model, and TXC groups. Real time-polymerase chain reaction was performed using a RT² Profiler™ PCR Array Rat ECM & Adhesion Molecules (PARN-013Z) (KangChen Bio-tech, China). The $2^{-\Delta\Delta Ct}$ was employed to calculate the fold change for each gene.

2.6. Confirmation of TXC efficacy

2.6.1. Enzyme linked immunosorbent assay (ELISA) analysis

ELISA kits (R&D Systems, USA) were used to measure IL-1 β , TNF- α , IL-6 and MMP 9 levels in the cell supernatant after treatment. Samples in 100 μ L were used for measurement by microplate spectrophotometry (Omega Bio-Tek, Inc., Norcross, GA, USA) at 405 nm.

2.6.2. Determination of potential mechanisms related to p38 MAPK pathway-mediated inflammation

Standard western blot and quantitative reverse transcription (qRT)-PCR assays were used to measure protein and mRNA expression of p38 MAPK pathway-mediated inflammation-related biomarkers.

Total RNA was extracted from chondrocytes using TRIzol (Invitrogen, USA). 1 μ g RNA was transcribed into cDNA using the RT kit (Thermo Fisher Scientific, Inc, USA), which was used for RT-PCR to determine mRNA expression with the SYBR Fluorescence Quantization Kit (Invitrogen) by a 7500 RT-qPCR System (Applied Biosystems, USA). The following primers were used: A disintegrin and metalloproteinase with thrombospondin motifs (ADAMTS) 4 forward 5'-GTC TCA GCA TTG ACC TTC CGT G-3' and reverse 5'-ACA GGG AGT TCC ATC TGC CAC C-3'; ADAMTS 5 forward 5'-GCA GAG TCC GAA CGA GTT TAC G-3' and reverse 5'-TAG TGC CAG TTC TGT GCG TCG-3'; Col1a1 forward 5'-CAT CGG TGG TAC TAA C-3' and reverse 5'-CTG GAT CAT ATT GCA CA-3'; Col5a1 forward 5'-ACA GCG GTC CCT GAC ACA CCT-3' and reverse 5'-GGG GCT GGG TCT GCT CCT CA-3'; MMP 3 forward 5'-ATG ATG ATG AAC GAT GGA CAG ATG-3' and reverse 5'-TGC TAC ACA TTG GTA AGG TCT CAG-3'; MMP 9 forward 5'-CGC TTG GAT AAC GAG TTC TCT G-3' and reverse 5'-TCA CAC GCC AGA AGT ATT TGT C-3'; MMP 13 forward 5'-GGA GTA ATC GCA TTG TGA GAG TC-3' and reverse 5'-GGT TCC AGC CAC GCA TAG TC-3'; Postn forward 5'-AGC ATC TTC CTC AGC CTC CT-3' and reverse 5'-TCC CCA ATC AGA ATC TCC CT-3'; and GAPDH forward 5'-ACG GCA AGT TCA ACG GCA CAG-3' and reverse 5'-GAA GAC GCC AGT AGA CTC CAC GAC-3'. Fold changes were calculated using formula $2^{-\Delta\Delta Ct}$ relative to expression in the sham group and glyceraldehyde-3-phosphate dehydrogenase (GAPDH) as the endogenous control.

Total protein was extracted from chondrocytes using radio-immunoprecipitation assay lysis buffer containing 1 mM phenylmethanesulfonyl fluoride (Beyotime Biotech, China); protein concentrations were measured by a bicinchoninic acid assay. Equal amounts of protein

abcam), p-P38 MAPK (4631; Cell Signaling Technology, Inc., USA), p38 MAPK (9212; Cell Signaling Technology, Inc., USA), and β -actin (ab8227; abcam, USA) overnight at 4 °C, followed by incubation with the horseradish peroxidase-conjugated secondary antibody IgG (ZB-2301; Zhongshan Goldenbridge Biotech, China) at room temperature. Data were quantitated using a Chemi Doc XRS+ (Bio-Rad, USA) and then normalized to the β -actin control.

Fig. 1. TXC delays cartilage and subchondral bone degeneration. A-F are representative images of HE staining, while G-I are representative micro-CT images. The left column is the sham group, the middle column is the model group, and the right column is the TXC-treated group. Normal pathological slices are shown as A (overview), D (trabecular arrangement of subchondral bone), and G (micro-CT image). Basic pathological changes of OA were observed in the model group, such as irregularity of chondrocyte arrangement (B, red arrow), destruction of cartilage surface (B, yellow arrow), and trabecular thickness of the subchondral bone (E, H). After TXC administration, pathological changes of the femoral condyle were significantly ameliorated, arrangement of chondrocytes became regular (C, red arrow), destruction of the surface and fibrillation of the cartilage was relieved (C, yellow arrow), and trabecular thickness of the subchondral bone improved (F, I). The regions of interest (ROIs) are the femoral condyle (red rectangle) and tibia plateau (yellow rectangle) in the subchondral bone (G-I), and the reconstructed ROIs are of the subchondral bone. J is the BMD of the femoral condyle, K is the BV/TV of the femoral condyle, L is the BMD of the tibia plateau, and M is the BV/TV of the tibia plateau. After TXC administration, the BMD and BV/TV of the femoral condyle significantly improved (J,K), while those of the tibia plateau exhibited no difference (L,M). * $P < 0.05$, ** $p < 0.01$. bar = 200 μm . (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

abcam) and MMP 3 (sc-21732; Santa Cruz Biotechnology) and an Alexa Fluor 488 goat anti-rabbit IgG (H + L) secondary antibody (R37116; Thermo Fisher Scientific, Inc., USA). Images were captured under a laser scanning confocal microscope (Olympus Corporation, Japan).

2.7. qRT-PCR assay for expression of microRNA-27b (miR-27b), miR-140, miR-92a-3p, and miR-34a

Expression of miR-27b, miR-140, miR-92a-3p, and miR-34a was evaluated by TaqMan microRNA assays, a form of qRT-PCR. Total RNA was extracted using a mirVana™ isolation kit (Invitrogen) as per manufacturer's instructions. RT was performed with miRNA-specific stem

loop RT primers and the TaqMan microRNA RT Kit (Applied Biosystems, USA). Expression of miR-27b, miR-140, miR-92a-3p, and miR-34a was detected using the TaqMan Universal PCR Master Kit (Applied Biosystems, USA). Target miRNAs fold changes were calculated relative to expression in untreated cells; U6 was used as an endogenous control.

2.8. Statistical analysis

SPSS software (version 17.0; SPSS Inc., USA) was used for statistical analysis. One-way analysis of variance was used, followed by the Student-Newman-Keuls *q* test with post-hoc correction for multiple

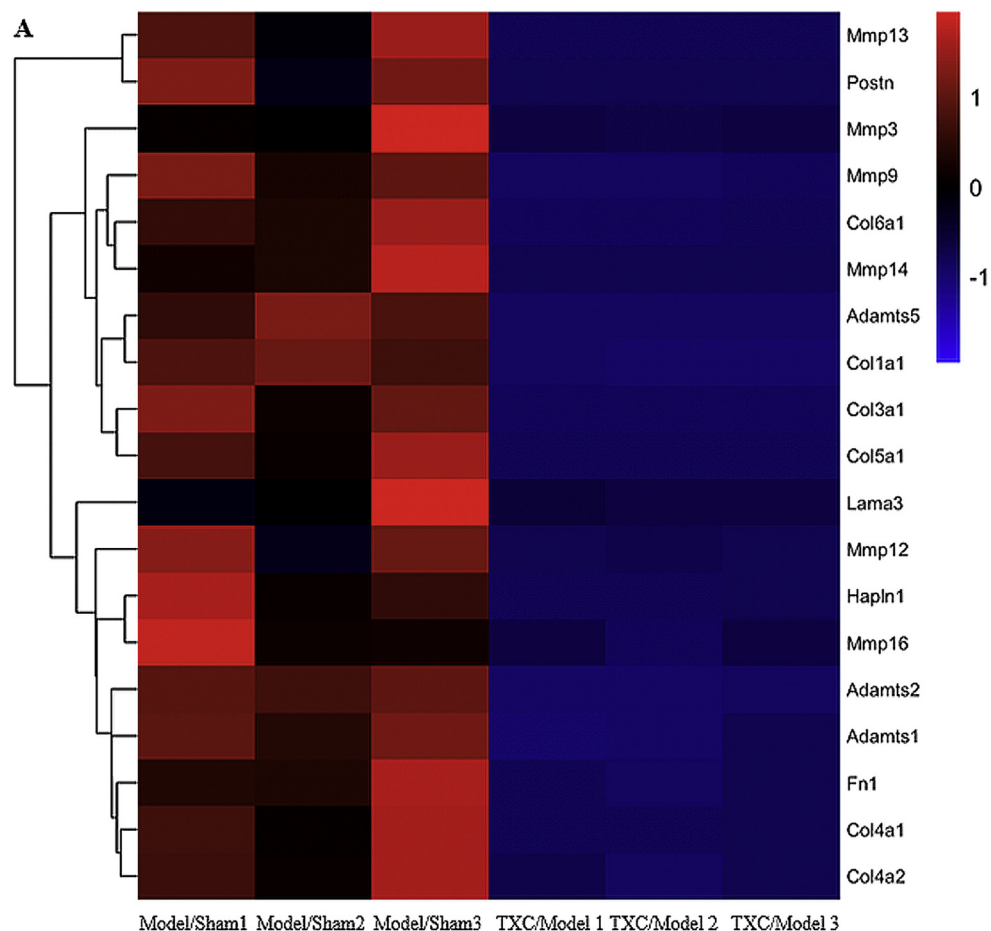
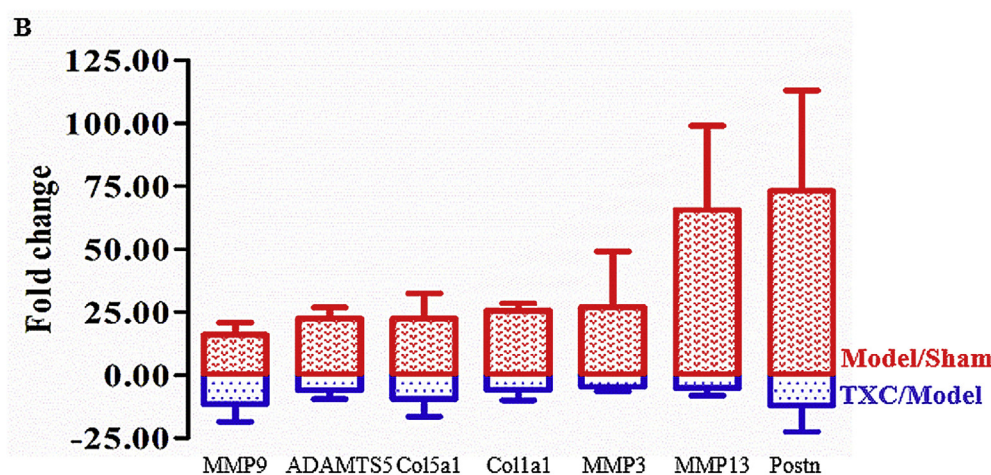


Fig. 2. Differential gene expression in TXC/model and model/sham. A. Heatmap of differential gene expression of knee articular cartilage samples. The total expression of 19 genes was higher (more than three-fold changes) in the model group (vs. sham group). However, only expression of MMP9, Adamts5, Col5a1, Col1a1, MMP3, MMP13, and Postn were significantly downregulated by TXC administration (more than four-fold change and $P < 0.05$). The seven identified genes were further investigated. B. Relative mRNA expression of the selected seven genes that fell into specific biological processes in OA (more than four-fold change) by TXC administration.



comparisons. Data from at least three independent experiments are presented as the mean $\pm$ standard deviation. A value of $p < 0.05$ was considered statistically significant.

3. Results

3.1. Confirmation of therapeutic effects of TXC on cartilage degeneration in the modified Hulth's OA rat model in vivo

3.1.1. TXC delays cartilage degeneration in OA rat model

In the model group, HE staining of the femoral condyle showed a rough cartilage surface with significant fibrosis, unclear structure and tidal lines of the cartilage layer, proliferation of chondrocytes, disorganized cell alignment, blood vessels passing through the tide line (Fig. 1B), and trabecular thickness of the subchondral bone (Fig. 1E). Micro-CT images of the tibia platform and femoral condyle of the model group revealed a rough surface of cartilage with focal surface fissures and flaking, formation of round or oval osteophytes, small and disorganized trabeculae, and cystic degeneration of the subchondral bone compared with the sham group (Fig. 1H). The BMD and BV/TV of the model group were lower than those of sham group (Fig. 1J–M). These pathological alterations confirm the successful establishment of the OA rat model.

After administration of TXC for 12 weeks, such pathological changes were significantly ameliorated (Fig. 1C, F, I). Further, the BMD and BV/TV of the femoral condyle subchondral bone were significantly enhanced compared to the model group (Fig. 1J and K). However, the BMD and BV/TV of the tibia platform subchondral bone did not show significant improvement (TXC vs. model group). These results indicate that TXC restored the thickness of the cartilage layer and size, arrangement of cancellous bone trabeculae, and inhibition of osteophytes.

3.1.2. TXC inhibits inflammatory mediated cartilage degeneration in the OA rat model

Differential gene expression between the model group and sham group was analyzed by the RT² Profiler™ PCR Array Rat ECM &

Adhesion Molecules. We found differential expression of 19 genes (more than a three-fold change) between the two groups, with the model group expressing these genes at significantly higher levels (Fig. 2A). After administration of TXC for 12 weeks, gene expression showed opposite trends; however, only expression of MMP 9, ADAMTS 5, Col5a1, Col1a1, MMP 3, MMP 13, and Postn were significantly lower (more than a four-fold change and $p < 0.05$, Fig. 2B) in the TXC group compared to the model group, which fall into the specific biological processes previously implicated in OA (Chinzei et al., 2018; Kim et al., 2014; Tsezou, 2014). Therefore, these seven genes were identified as the potential targets of the therapeutic effects of TXC on cartilage degeneration and were selected for further investigation by western blot and RT-PCR.

As shown in Fig. 3, all seven genes were significantly upregulated in the model group and downregulated in TXC-treated group (Fig. 3). These results suggest that ADAMTS 5, Col5a1, Col1a1, MMP 3, MMP 9, MMP 13, and Postn may play roles in the mechanisms of TXC's therapeutic effects.

3.2. Evidence that TXC's effects may be via suppression of inflammation-mediated cartilage degeneration in LPS-exposed chondrocytes in vitro

3.2.1. TXC decreases inflammatory cytokines in LPS-exposed chondrocytes

After administration of TXC and SB203580 for 8 h, only large doses (300 μ g/mL, 400 μ g/mL) of TXC and 10 μ M SB203580, levels of IL-1 β and TNF- α in culture solution decreased markedly, despite all TXC doses causing a significant decrease ($p < 0.05$, Fig. 4A and B). Therefore, we selected 300 μ g/mL and 8 h TXC treatment for subsequent experiments.

After administration of 300 μ g/mL TXC and 10 μ M SB203580 for 8 h, the concentration of IL-6 and MMP 9 in culture solution was significantly lower than in the LPS-exposed cells (Fig. 4C and D). These results suggest that TXC's therapeutic effect may relate to decreasing inflammatory cytokines via the p38 MAPK pathway.

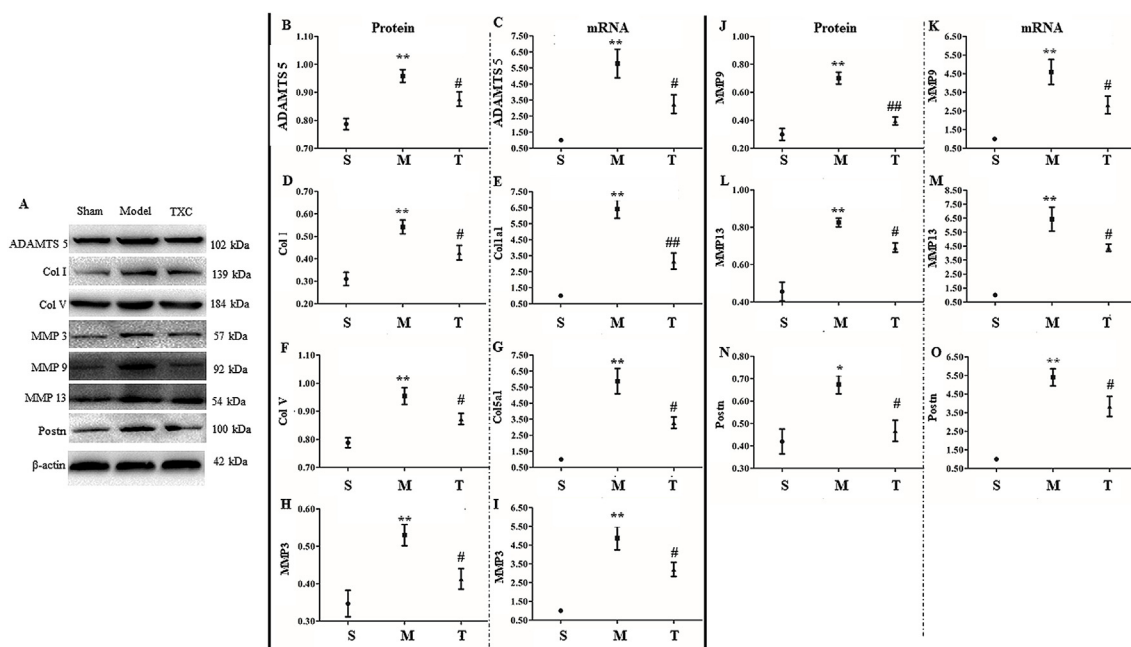


Fig. 3. Expression of ADAMTS 5, Col I, Col V, MMP 3, MMP 9, MMP 13, and Postn. A. Representative images of protein expression levels of ADAMTS5, Col I, Col V, MMP3, MMP9, MMP13, and Postn by western blot. B–O. Quantitative analyses of protein expression of these genes. The left column shows the results of relative protein expression, while the right column shows the results of mRNA expression. All results were analogous: gene expression was significantly upregulated in the model group, whereas administration of TXC significantly downregulated gene expression. S = sham group; M = model group; T = TXC-treated group. Sham vs. model: * $p < 0.05$; ** $p < 0.01$. Model vs. TXC: # $p < 0.05$; ## $p < 0.01$.

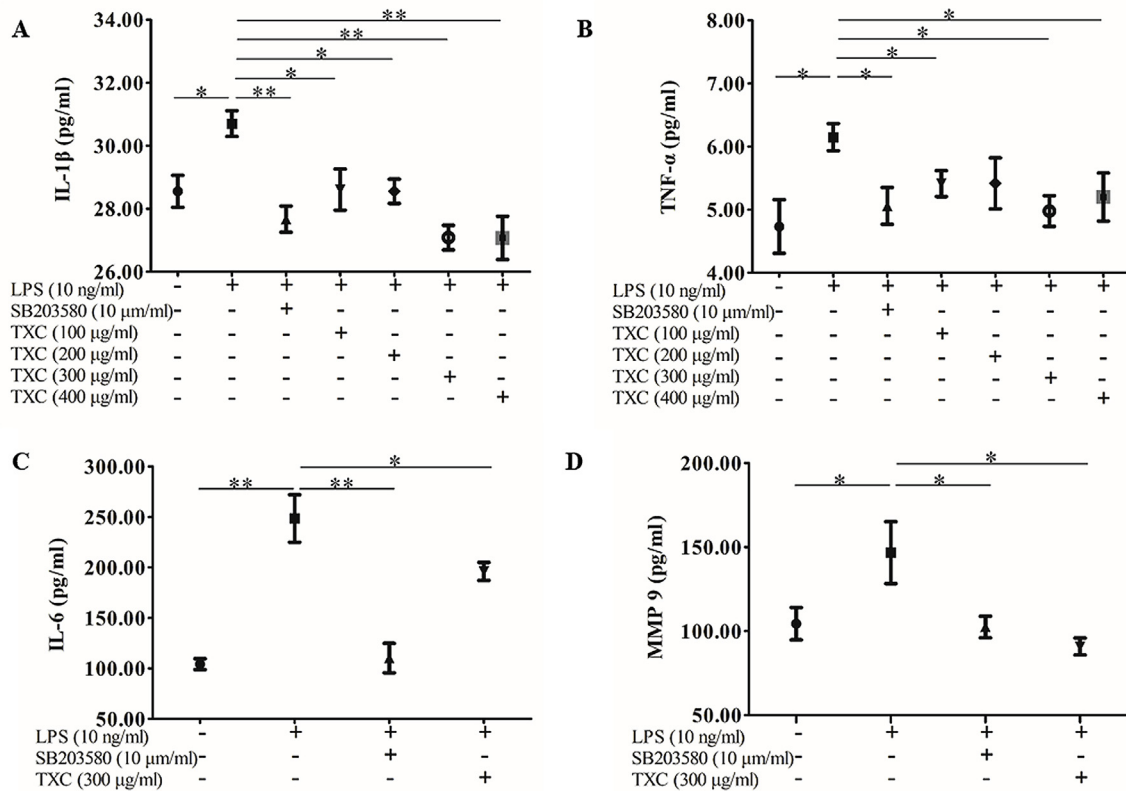


Fig. 4. TXC decreases inflammatory cytokines in LPS-exposed chondrocytes. TXC administration significantly reduced IL-1 β (A) and TNF- α (B) concentrations in culture solution for different doses. A dose of 300 μ g/mL TXC achieved the maximum effect. TXC administration at 300 μ g/mL significantly alleviated IL-6 and MMP 9 levels in culture solution. * $p < 0.05$; ** $p < 0.01$.

3.2.2. TXC reduces biomarker expression of p38 MAPK pathway-mediated inflammation

Expression of protein (Fig. 5A, C, E, G) and mRNA (Fig. 5B, D, F, H) of the biomarkers (ADAMTS 4, ADAMTS 5, MMP 3, and MMP 13) showed similar trends, i.e., upregulated by LPS exposure and downregulated by treatment with TXC and SB203580 (Fig. 5).

By contrast, the p-p38 MAPK/p38 MAPK ratio was upregulated by LPS exposure, but downregulated by TXC and SB203580 (Fig. 6).

Taken together, these results suggest that reduced biomarker expression via suppression of the p38 MAPK pathway may play a role in the mechanisms of the effects of TXC.

3.2.3. TXC affects expression of miRNAs in LPS-exposed chondrocytes

The results of the TaqMan microRNA assay showed that expression of miR-27b, miR-140, and miR-92a-3p in LPS-exposed cells was downregulated by LPS exposure and upregulated by treatment with SB203580 and TXC (Fig. 7A, B, C); expression of miR-34a showed the opposite tendency (Fig. 7D).

These results confirmed that suppression of p38 MAPK pathway-mediated inflammation plays a crucial role in the mechanisms of the effects of TXC, indicating that TXC has multi-target effects for treatment of complex diseases such as OA.

4. Discussion

In the present study, we confirmed TXC's protective effects in vivo (using an animal model) and in vitro (using LPS-exposed chondrocytes) on inflammation-mediated cartilage degeneration. Cartilage degeneration was closely associated with two mechanisms, namely inhibited inflammation and reduced ECM degeneration, since expression of miR-27b, miR-140, and miR-92a-3p was upregulated, while expression of miR-34a was downregulated. We also found evidence that the p-38

MAPK pathway is involved in TXC's therapeutic effects since expression of p-p38 MAPK was downregulated following TXC administration. The p38 MAPK pathway-related can, therefore, be regarded as one of the many complex mechanisms involved in the therapeutic effects of TXC, along with the ER stress pathway verified in our previous work (Liang et al., 2018). To our knowledge, this is the first report verifying the cartilage protection and p38 MAPK pathway-related mechanisms of TXC. This study thus provides evidence of the efficacy and mechanisms of herbal treatment of OA.

In the present study, we first employed a modified Hulth's method, which is considered the classic OA model (Wang et al., 2017, 2018). Pathological examination of our model revealed typical manifestations of OA, including a rough cartilage surface with significant fibrosis, unclear structure and tidal line of the cartilage layer, proliferation of chondrocytes, disorganized cell alignment, and blood vessels passing through the tide line (Fig. 1). Additionally, we found 19 highly expressed genes, which is consistent with previous work (Heinegard and Saxne, 2011). These findings confirmed our successful establishment of the OA model. We then administrated TXC to assess its effects. TXC was found to delay cartilage and subchondral bone degeneration (Fig. 1C, F, I), and BMD and BV/TV of the femoral condyle significantly improved. Moreover, cartilage degeneration related biomarkers, including ADAMTS 5, Col I, Col V, MMP 3, MMP 9, MMP 13, and Postn, were significantly suppressed by TXC (Figs. 2 and 3). Enhanced catabolic response can exacerbate cartilage matrix degeneration by promoting matrix-degrading genes like ADAMTS and MMPs (Heinegard and Saxne, 2011), whereas this study found that TXC may downregulate the expression of these genes. Thus, cartilage protection by TXC was verified by pathological and genetic data.

Next, we employed a classical LPS-exposed chondrocyte model to simulate an inflammatory and cartilage degenerative system as an OA cell model. We found that IL-1 β and TNF- α were significantly increased

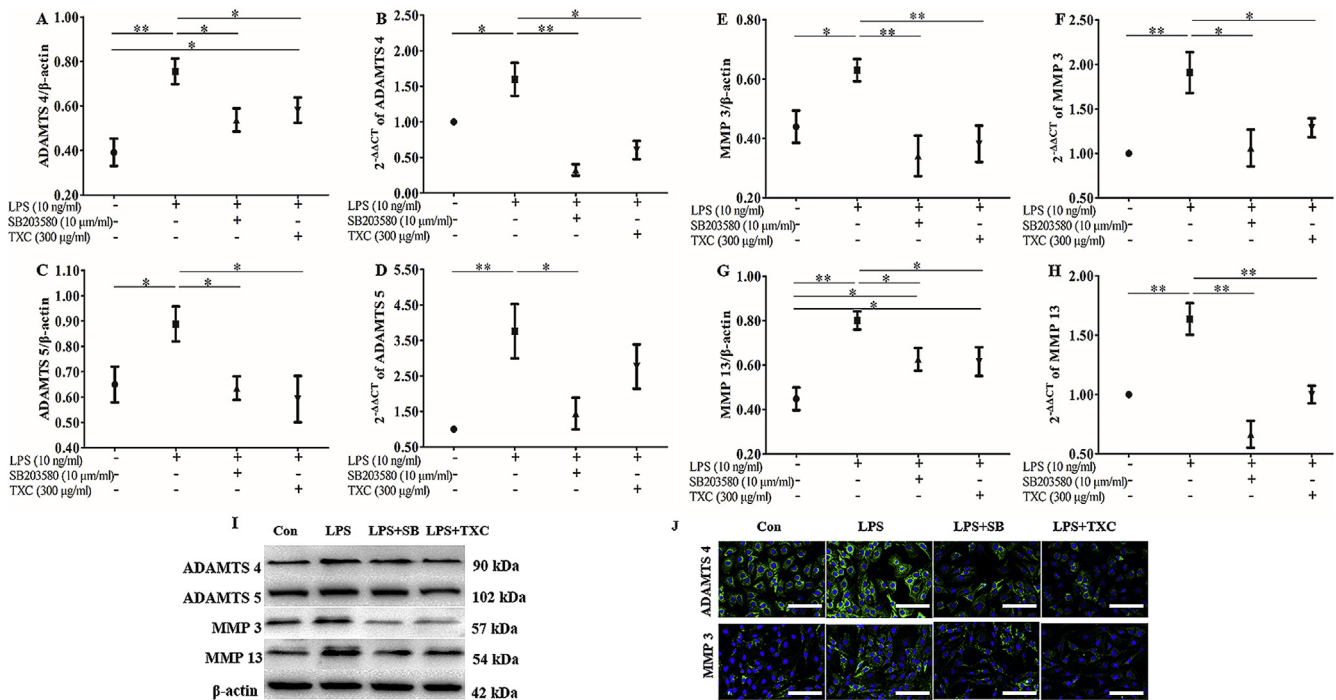


Fig. 5. TXC suppressed expression of ADAMTS 4, ADAMTS 5, MMP 3, and MMP 13 in LPS-exposed chondrocytes. A-H. The left column shows the protein expression results, while the right column shows the relative mRNA expression results. All results are analogous. Exposure to LPS significantly upregulated the expression of ADAMTS 4, ADAMTS 5, MMP 3, and MMP 13. Treatment with SB203580 and TXC significantly suppressed expression of ADAMTS 4, ADAMTS 5, MMP 3, and MMP 13. Treatment with TXC significantly suppressed mRNA expression of MMP 13, whereas inhibition of protein expression of MMP 13 did not reach statistical significance. I. Representative images of protein levels of ADAMTS 4, ADAMTS 5, MMP 3, and MMP 13 by western blot analyses. J. Representative images of immunohistochemical staining. Green cells indicate ADAMTS 4 + and MMP 3 + cells. ADAMTS 4 + and MMP 3 + cells were enhanced by LPS exposure and treatments by LPS + TXC. Intact cells vs. LPS-exposed cells: *p < 0.05; **p < 0.01. LPS-exposed vs. treatments by SB203580 and TXC: #p < 0.05; ##p < 0.01. LPS-exposed vs. treatment by TXC: Δp < 0.05; ΔΔp < 0.01, bar = 200 μm. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

in LPS-exposed chondrocytes, confirming the successful establishment of this cell model (Fig. 4A and B). Our results showed that IL-1β, TNF-α, IL-6, and MMP 9 significantly increased in LPS-exposed chondrocytes, while administration of TXC, as well as SB203580, decreased these enhancements (Fig. 4). These results were in agreement with the findings using a rabbit OA model in our previous study (Wu et al., 2019). Reproducing the similar results in different animal models proved the reliability of the present experimental system. Moreover, expression of ADAMTS 4, ADAMTS 5, MMP 3, and MMP 13 displayed similar results. LPS exposure may upregulate expression of these genes, while TXC administration and SB203580 may significantly suppress this upregulation (Fig. 5). The roles of inflammatory cytokines, the MMPs family, and the ADAMTS family in OA pathogenesis have been well documented. Among the ADAMTS family, ADAMTS 4 and ADAMTS 5 have been considered as the major aggrecanases involved in cartilage matrix turnover. IL-1β, IL-6, and TNF-α can induce the production of MMPs and ADAMTS and inhibit the synthesis of proteoglycans and Col II,

which play pivotal roles in cartilage matrix degeneration and bone resorption in OA. MMP is a family of proteases mainly associated with tissue remodeling and degeneration of ECM, particularly MMP 3, MMP 9, and MMP 13, and may cause degeneration Col II and glycans in ECM, resulting in OA cartilage destruction (Murphy and Nagase, 2011). Moreover, chondrocytes can release several aggrecanases like ADAMTS 4 and ADAMTS 5, which play a crucial role in OA pathophysiology without participation of MMPs (Malfait et al., 2002). LPS-exposed chondrocytes release more MMP 3, MMP 13, ADAMTS 4, and ADAMTS 5, and reducing them is protective (Ross et al., 2012; Scotece et al., 2018; Sun et al., 2006). Our results were in accordance with these findings. We also verified TXC's therapeutic efficacy at a cellular level.

Interestingly, SB203580, an antagonist of the p38 MAPK pathway, exhibited a synergistic effect with TXC, which strongly suggests a role of the p38 MAPK pathway in the pathogenesis of OA and efficacy of TXC. It has been reported that the p38 MAPK pathway plays a significant role in the progression of OA by leading to overexpression of

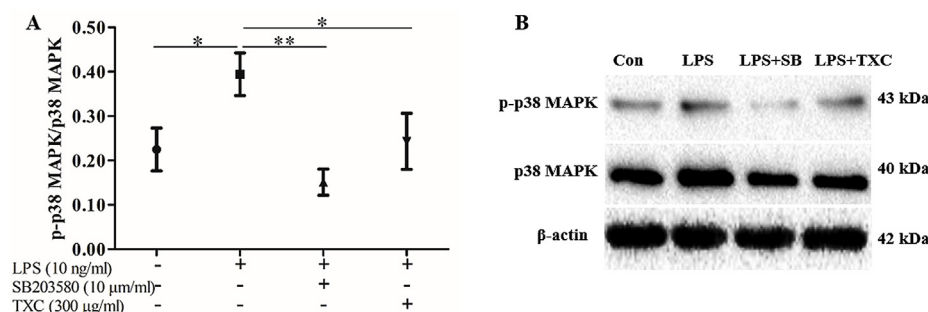


Fig. 6. TXC decreased p-p38 MAPK expression in LPS-exposed chondrocytes. A. LPS exposure significantly enhanced the ratio of p-p38 MAPK/p38 MAPK. Treatment with SB203580 + TXC and treatment with TXC significantly reduced this ratio. B. Representative images of p-p38 MAPK and p38 MAPK by western blot analysis. *p < 0.05; **p < 0.01.

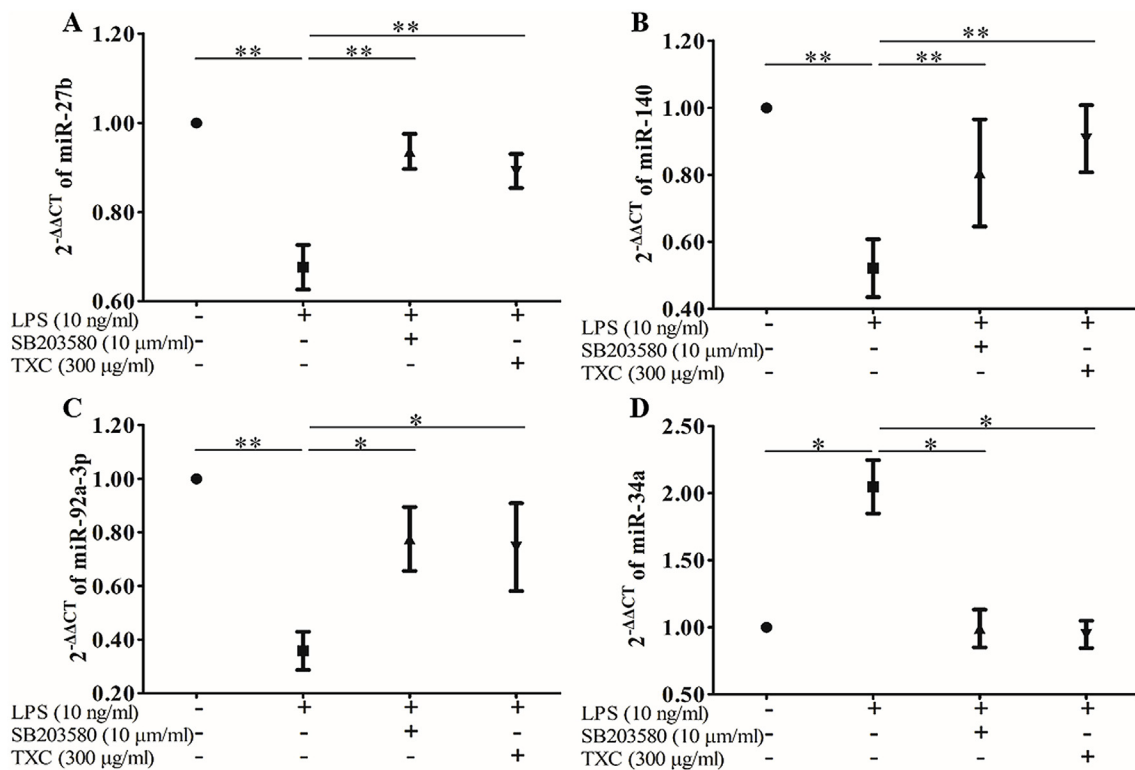


Fig. 7. TXC treatment affects expression of miRNAs in LPS-exposed chondrocytes. LPS exposure significantly downregulated the expression of miR-27b (A), miR-140 (B), and miR-92a-3p (C), but upregulated the expression of miR-34a (D). Treatments with SB203580 and TXC significantly upregulated the expression of miR-27b (A), miR-140 (B), and miR-92a-3p (C), but downregulated the expression of miR-34a (D) in LPS-exposed chondrocytes. **p* < 0.05; ***p* < 0.01.

MMP 3, MMP 13, ADAMTS 4, and ADAMTS 5 in osteoarthritis chondrocytes (Park et al., 2018; Sondergaard et al., 2010; Yaykasli et al., 2015; Yu and Kim, 2015). Another report indicated that MAPKs/NF-κB pathways can activate the inflammation response in human primary OA chondrocytes (Scotece et al., 2018). One study confirmed that Geniposide (GE), an iridoid glycoside purified from the Chinese herb *Gardenia jasminoides* Ellis (Rubiaceae), may deliver its anti-inflammatory effects by suppressing activation of the p38 MAPK pathway (Chen et al., 2018). Taken together, these findings support that the p38 MAPK pathway plays a role in the pathogenesis of OA. In the present study, we found that TXC decreased the ratio of p-p38 MAPK/p38 MAPK enhanced by LPS exposure (Fig. 6), strongly suggesting the role of the p38 MAPK pathway in the TXC effects, which warrants further investigation.

Our miRNA results showed that expression of miR-27b, miR-140, and miR-92a-3p were significantly downregulated in LPS-exposed cells, but upregulated by TXC and SB203580 (Fig. 7A, B, and C); in contrast, expression of miR-34a was significantly upregulated in LPS-exposed cells and downregulated by TXC and SB203580 (Fig. 7D). The network and interactions of miRNAs, such as miR-27b, miR-140, miR-92a-3p, and miR-34a, are quite complicated and they play roles as regulators of pro-inflammatory cytokines and proteinase gene expression. For example, miR-27b expression is lower in IL-1β stimulated chondrocytes that enhance the production of MMP 13 (Akhtar et al., 2010); miR-140 is expressed specifically in cartilage and regulates ECM-degrading enzymes, which modulates ECM homeostasis to alleviate OA progression by promoting Col II expression and inhibiting MMP 13 and ADAMTS 5 expression (Si et al., 2017); miR-92a-3p shows significantly lower expression in OA cartilage, which enhances H3 acetylation on aggrecan, cartilage oligomeric protein (COMP) and Col2a1 promoter, and promotes relative cartilage matrix expression; and miR-92a-3p downregulates ADAMTS-4/5 expression and suppresses the activation of MAPK and NF-κB in IL-1β-induced chondrocytes (Mao et al., 2017a,

2017b); and miR-34a is thought to contribute to cell apoptosis or cell cycle arrest by suppressing miR-34a expression rescue IL-1β-induced Cyr61 suppression and promoting OA chondrocyte proliferation through ADAMTS 4 (Yang et al., 2018). In this regard, upregulation of miR-27b, miR-140, and miR-92a-3p, along with downregulation of miR-34a, is thought to be protective; by contrast, downregulation of miR-27b, miR-140, and miR-92a-3p, along with upregulation of miR-34a, is harmful. Our results proved that TXC and SB203580 are protective for OA cells, in which the p38 MAPK pathway may play a role.

Given our previous work on the ER stress pathway (Liang et al., 2018), we believe that OA mechanisms are quite complicated. It is likely that the ER stress pathway and p38 MAPK pathway are only part of the picture. An imbalance in gene expression skewing inflammatory pathways can cause aberrant turnover and loss of ECM proteins in articular cartilage during OA progress. Interactions among such pathways must be fully considered in OA or TXC research. Our findings indicated that OA is an inflammatory disease. We thus believe that targeting inflammatory mechanisms may contribute to developing new transformative therapies, which is in accordance with previous work (Li et al., 2018; Utomo et al., 2016).

Another important issue is the toxic effect of TXC. There is no clinical report investigating the toxicity of TXC. Yu et al. performed the subacute toxicity test for TXC. They did not find any changes of blood routine examination, blood biochemical examination and cytochrome P450 expression in SD rats after administration of the TXC up-dosing to 20, 40 and 80 times the clinically recommended dose (1840 mg/d) (Yu et al., 2014). Another acute toxicity study found that the fatal dose is 525.7 times the clinical dose, and the dose of 366.1 times the clinical dose is still safe (Liao et al., 2012). Both studies therefore concluded that TXC is a safety herbal compound. But some reports mentioned the side-effects of the component herbs of TXC in a large dose (Liao et al., 2012). We believe that the toxicity, especially the cytotoxicity of TXC needs further investigation, which will be included in our future study.

5. Conclusion

The present study confirmed the therapeutic effects of TXC in animal and cellular OA models. We also found that p38 MAPK pathway-related mechanisms might play a role in cartilage protection by TXC. The findings of this study and our previous work support that OA is an inflammatory disease, and thus targeting inflammatory mechanisms may contribute to the development of new therapies for OA.

Authors' contributions

XL, HY and TA got the original idea. XL, ZZ, WL, JZ, XS, LX, LJ, XH, HL, CZ, HY, and TA performed the experiments and analyzed the data. XL, HY and TA ran the statistics. XL and TA wrote the draft. XL, ZZ, WL, JZ, XS, LX, LJ, XH, HL, CZ, HY, and TA discussed and approved the final manuscript. HY and TA supervised the study.

Declaration of competing interest

The authors declared no conflict of interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jep.2019.112390>.

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